

CA1 Hippocampal Pyramidal Neuron — NEURON Simulation Parameters

Biological parameters translated into NEURON simulator units for baseline control simulation. All 10 available input types included. Terminology mapping, unit conversions, compartment mapping, and synapse count scaling fully documented. Option A — single Exp2Syn per synapse type.

Model: ModelDB 2018009 (CA1_PC_cAC_sig) · Morphology: mpg141017_a1-2_idC.asc · Celsius = 34 · v_init = -70 mV

All synapse counts at 10% of Bezaire & Soltesz (2013) per-cell estimates — proportional scaling preserves E/I ratio · Dr. Cavarretta Lab

Table 1A — Terminology and Unit Conversion Reference

Maps every term from the biological parameter table to the corresponding NEURON simulation term, shows the unit in each context, and gives the conversion formula. Peak conductance (gmax) in the biological table becomes gmax/weight in NEURON. All other parameters are used directly without conversion except firing frequency. Blue cells show the conversion formula.

BIOLOGICAL TABLE TERM	BIOLOGICAL UNIT	→	NEURON TABLE TERM	NEURON UNIT	CONVERSION FORMULA
Peak Conductance (gmax)	nS	→	gmax / weight	μS	Biological value ÷ 1000
Synaptic rise time	ms	→	tau1	ms	No conversion — used directly
Synaptic decay time	ms	→	tau2	ms	No conversion — used directly
Reversal potential (excitatory)	mV	→	e	mV	No conversion — 0 mV for AMPA
Reversal potential (inhibitory)	mV	→	e	mV	No conversion — -70 to -75 mV for GABA-A
Firing frequency	Hz	→	NetStim interval	ms	1000 ÷ Hz
Temperature	°C	→	celsius	°C variable	No conversion
Initial membrane potential	mV	→	v_init	mV variable	No conversion
EPSC amplitude (NR only)	pA at holding Vm	→	gmax / weight	μS	$g = \text{EPSC} \div (V_m - E_{\text{rev}}) \div 1000$
Synapse count (all inputs)	Per-cell count (Bezaire & Soltesz 2013)	→	N synapses	10% baseline count	Full count × 0.10

Table 1B — Biological Layer to Model Compartment Mapping

Maps each hippocampal layer to exact section names in the CA1_PC_cAC_sig model. Distances from soma[0](0.5) confirmed by NEURON h.distance() query across all 189 apical and 44 basal sections. Section assignment determines where each synapse type is placed in the simulation.

BIOLOGICAL LAYER	DISTANCE FROM SOMA	MODEL SECTIONS (NEURON)	TOTAL SECTIONS	SYNAPTIC INPUTS PLACED HERE
Stratum oriens (SO)	0–300 μm (basal)	dend[0]–dend[43]	44	CA3 (CA3a), Bistratified, Ivy cells
Stratum pyramidale	0 μm (soma)	soma[0]	1	PV+ Basket cells
SR proximal	0–150 μm (apical)	apic[] dist 0–150 μm apic[0]–apic[8] + scattered	61	CA3 Schaffer collaterals (proximal)
SR distal	150–300 μm (apical)	apic[] dist 150–300 μm apic[9]–apic[18] + scattered	82	CA3 Schaffer collaterals (distal), Bistratified
Stratum lacunosum-moleculare (SLM)	300+ μm (apical)	apic[] dist > 300 μm apic[19–28], apic[126–149] and others	46	Entorhinal Cortex, Nucleus Reunions, O-LM, Neurogliaform

Methods note for Table 1B:

Section counts and layer assignments were determined by running NEURON's `h.distance()` function from `soma[0](0.5)` across all 189 apical (`apic[]`) and 44 basal (`dend[]`) sections of morphology file `mpg141017_a1-2_idC.asc`. Each section was assigned to a layer based on the distance of its midpoint (0.5 position) from the soma. Layer boundaries follow published CA1 anatomy: SO = all basal dendrites (`dend[]`); SR proximal = apical sections where midpoint distance 0–150 μm ; SR distal = apical sections where midpoint distance 150–300 μm ; SLM = apical sections where midpoint distance >300 μm . Boundaries consistent with Megias et al. (2001) PMID:11226691 and Klausberger & Somogyi (2008) PMID:18599766.

Section counts are exact values confirmed from this specific morphology file. They will differ for other morphology reconstructions. The distance filter is applied in simulation code using: `h.distance(seg.x, sec=section)` evaluated at each segment midpoint.

Table 2 — NEURON Simulation Parameters (Baseline Control)

Complete parameter set for all 10 included input types. gmax/weight column shows the value entered into NEURON NetCon — this is the peak conductance in μS . Bracketed numbers [1]–[6] refer to derivation notes below the table. Full Count = Bezaire & Soltesz (2013) per-cell estimate. 10% Baseline = synapse count used in simulation.

INPUT SOURCE	MECHANISM	TARGET SECTIONS	TAU1 (MS)	TAU2 (MS)	E (MV)	GMAX /WEIGHT (MS)	NETSTIM INTERVAL (MS)	FULL COUNT	10%B ASELINE	SOURCE
► Excitatory Inputs — Glutamatergic										
CA3 Schaffer Collaterals (proximal) Hippocampal CA3 → CA1	Exp2Syn	apic[] dist 0–150 μm apic[0]–apic[8] + scattered	0.5	3.0	0	0.00256 [1] 2.56 nS $\div$ 1000	100 ms (10 Hz)	~1,800	180	Andrásfalvy & Magee 2001 [9] Bezaire & Soltesz 2013 [7]
CA3 Schaffer Collaterals (distal) Distance-scaled — 3 $\times$ proximal	Exp2Syn	apic[] dist 150–300 μm apic[9]–apic[18] + scattered	0.5	3.0	0	0.00796 [1] 7.96 nS $\div$ 1000	100 ms (10 Hz)	~1,800	180	Andrásfalvy & Magee 2001 [9] Nicholson et al. 2006 [10]
Entorhinal Cortex Layer III Temporoammonic pathway	Exp2Syn	apic[] dist > 300 μm SLM — co-targeted with O-LM	0.5 [assumed]	3.0 [assumed]	0	0.00150 [2] working estimate — ~60% of CA3 proximal	1000 ms (~1 Hz — sparse firing; median 0.4 Hz in vivo, range 0.4–2.1 Hz)	~200	20	Nicholson et al. 2006 [10] Bezaire & Soltesz 2013 [7]
Nucleus Reunions Thalamic midline — mPFC $\leftrightarrow$ RE $\leftrightarrow$ CA1	Exp2Syn	apic[] dist > 300 μm SLM — co-targeted with entorhinal	0.5 [assumed]	3.0 [assumed]	0	0.00134 [3] 86.8 pA $\div$ 65 mV = 1.34 nS $\div$ 1000	200 ms (theta ~5 Hz)	~50	5	Goswamee et al. 2021 [8] Bezaire & Soltesz 2013 [7]
► Inhibitory Inputs — GABAergic										
PV+ Basket Cells Parvalbumin-expressing; fast-spiking	Exp2Syn	soma[0] Perisomatic — soma & proximal dendrites	4.6	32.4	–75	0.00095 [4] 0.95 nS $\div$ 1000	45.5 ms (22 Hz theta)	183	18	Buhl et al. 1995 [12] Klausberger et al. 2003 [3] Bezaire & Soltesz 2013 [7]
Bistratified Cells	Exp2Syn	apic[] 50–300 μm AND dend[] 0–300 μm	2.0	16.1	–70	0.00099 [5] 0.86 mV $\div$ 14.4	169.5 ms (5.9 Hz theta)	104	10	Pawelzik et al. 1999 [18] Maccaferri et al. 2000 [11]

INPUT SOURCE	MECHANISM	TARGET SECTIONS	TAU1(MS)	TAU2(MS)	E(MV)	GMAX /WEIGHT (MS)	NETSTIM INTERVAL (MS)	FULL COUNT	10%B ASELINE	SOURCE
PV+, SOM+, NPY+; dendrite-targeting		Co-aligned with CA3 input in SR and SO				mV × (1/60.2 MΩ) = 0.99 nS ÷ 1000				Bezaire & Soltesz 2013 [7]
O-LM Interneurons Oriens-lacunosum moleculare; SOM+	Exp2Syn	apic[] dist > 300 μm SLM — co-targeted with entorhinal	6.2	20.8	-70	0.00037 [4] 26 pA ÷ 70 mV = 0.37 nS ÷ 1000	76.9 ms (~13 Hz theta)	77	8	Maccaferri et al. 2000 [11] Klausberger et al. 2003 [3] Bezaire & Soltesz 2013 [7]
Ivy Cells nNOS+, NPY+; most abundant interneuron	Exp2Syn	apic[] 0–300 μm AND dend[] 0–300 μm Proximal basal and oblique dendrites	2.8 (mean 2.6–3.0 ms)	16.8 (mean 15.8–17.8 ms)	-70	0.000115 [4] midpoint 0.09–0.14 nS ÷ 1000	1428.6 ms (0.7 Hz theta)	422	42	Fuentealba et al. 2008 [5] Bezaire & Soltesz 2013 [7]
Neurogliaform Cells α-actinin-2+, COUP-TFII+, reelin+	Exp2Syn	apic[] dist > 300 μm SLM — dense local axonal plexus	2.8 [assumed ivy]	16.8 [assumed ivy]	-70	0.000115 [assumed ivy — no direct value]	150 ms (~6.7 Hz theta mean)	140	14	Fuentealba et al. 2010 [6] Armstrong et al. 2012 [14] Bezaire & Soltesz 2013 [7]

Derivation Notes for gmax / weight column:

[1] CA3 AMPA conductance — directly measured by Andrásfalvy & Magee (2001) [9] using outside-out patch recordings excised directly from the apical dendrite at defined distances from soma. Outside-out patches physically remove the membrane from the intact neuron — A-type potassium channels are absent during measurement. Therefore these values represent pure synaptic conductance at the synapse location, not somatic recordings after dendritic filtering. A-type channel attenuation is handled separately by the distance-dependent κ channel distribution already built into the model (cellM1.hoc). No correction for dendritic filtering is required and no double attenuation occurs. The ~3× increase in conductance from proximal (2.56 nS) to distal (7.96 nS) reflects a genuine biological increase in AMPA receptor number at distal synapses, as confirmed by Magee & Cook (2000) [22] who showed somatic EPSP amplitude is location-independent due to this scaling.

[2] Entorhinal conductance and firing rate — conductance is a working estimate at ~60% of CA3 proximal based on Nicholson et al. (2006) showing AMPA receptor expression in SLM is substantially lower than SR. No direct nS measurement available. NetStim interval set to 1000 ms (~1 Hz) based on in vivo recordings showing entorhinal layer III neurons fire very sparsely: median 0.4 Hz in freely moving rats (Burgalossi et al. 2014 [20]) and mean persistent firing of 2.1 Hz range 0.4–5.5 Hz (Tahvildari & Alonso 2005 [21]). Both values to be refined during baseline validation.

[3] Nucleus Reunians conductance — derived from EPSC amplitude: $g = I \div (V_m - E_{rev}) \div 1000 = 86.8 \text{ pA} \div (-65 - (-130) \text{ mV}) \div 1000 = 1.34 \text{ nS} \div 1000$. Holding potential -65 mV, assumed reversal -0 mV for AMPA. Goswamee et al. (2021).

[4] Conductance converted from current clamp IPSP or IPSC amplitudes using $g = \text{amplitude} \div (V_m - E_{rev}) \div R_{in}$ where applicable, or direct voltage clamp conversion.

[5] Bistratified conductance — directly derived from Pawelzik et al. (1999) CA1 data: IPSP amplitude $0.86 \pm 0.55 \text{ mV}$ at $V_m -57.6 \text{ mV}$, $E_{rev} -72 \text{ mV}$ (from paper), $R_{in} 60.2 \text{ M}\Omega$ (Buhl et al. 1996). $g = 0.86 \div (-57.6 - (-72)) \times (1/60.2) = 0.86 \div 14.4 \times 0.0166 = 0.00099 \mu\text{S}$. $n=18$ bistratified cells, adult rat CA1.

Table 3 — Exclusions, Simplifications, and Future Steps

All conscious methodological decisions documented with scientific justification. CA2 and septal inputs excluded entirely. NMDA excluded from baseline only — highest priority addition before seizure experiments. All other exclusions reflect limitations of the Exp2Syn mechanism.

EXCLUDED ELEMENT	CATEGORY	REASON FOR EXCLUSION	FUTURE PLAN
CA2 inputs	Excitatory input	No quantitative parameters available for CA2→CA1 synaptic conductance, synapse counts, or firing frequency. Chevaleyre & Siegelbaum (2010) characterizes pathway at population level only — no unitary patch clamp data.	Monitor literature. Exclude from all simulations until unitary parameters published.
Septal GABAergic inputs	Inhibitory input	Bezaire & Soltesz (2013) indicates septal GABAergic afferents primarily target interneurons not pyramidal cells directly. Contribution to pyramidal cell inhibition is minor and unquantified.	Exclude from single-cell model. Relevant for network-level theta pacemaking only.
NMDA receptors	Synapse mechanism	Requires separate .mod file implementing Jahr & Stevens (1990) Mg2+ voltage-dependent block. Exp2Syn cannot model voltage-dependent gating. NMDA is critical for seizure modeling — sustained depolarization removes Mg2+ block and drives ictal activity.	HIGHEST PRIORITY. Add NMDA.mod before seizure experiments begin.
CCK+ Basket Cells	Inhibitory input	No direct unitary conductance measurement available for CCK+ basket cells in CA1. Hefft & Jonas (2005) provides quantal amplitude (33.1 ± 3.9 pA) in dentate gyrus granule cells only — not CA1 pyramidal cells. All kinetic parameters (τ_1 , τ_2) would require assumption from PV basket cells. Synapse count (106) and location are known but the core simulation parameter — conductance — has no direct CA1 source.	Search for paired recording data from identified CCK+ basket cells onto CA1 pyramidal cells. Add when direct CA1 conductance value becomes available.
10% synapse count scaling	Baseline simplification	All synapse counts at 10% of Bezaire & Soltesz (2013) per-cell estimates. E/I ratio preserved by consistent proportional scaling. Reduces computational load and facilitates stepwise validation.	Scale to full counts after baseline validation confirms correct E/I balance and realistic firing behavior.

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